

Engineering Change Order ECO-2024-067

Product: ACT-PKR-4500 'Patriot' Retrievable Packer | Revision B → C

ECO Number / Date	ECO-2024-067 / 2024-10-15
Originator	R. Callahan
Affected Product	ACT-PKR-4500 'Patriot' Retrievable Packer
Revision Change	B to C
Change Classification	Class 2 (option addition - no form/fit change to std product)
Effectivity	Serial numbers P45-24-0900 and later may be redressed with -03-AF kit.

1. Reason for Change

Operators in West Texas CO2 flood service reported HNBR element hardening after 18+ months at 8-12% CO2 with intermittent H2S. Aflas (TFE/P) offers superior resistance to amine inhibitors, CO2 and H2S at up to 350 degF.

2. Description of Change

Add optional Aflas element kit ACT-PKR-4500-03-AF (center element 90A compound ACT-A90, end elements 85A compound ACT-A85) as an orderable alternative to the standard HNBR element assembly ACT-PKR-4500-03. Standard product is unchanged. Datasheet temperature rating with Aflas option: 350 degF. O-rings in kit remain Viton 75A (compatible).

3. Affected Documents

Document	From Rev	To Rev	Change Summary
BOM-PKR-4500	B	C	Add -03-AF optional kit; note on item 4
DS-PKR-4500	B	C	Add Aflas option to specs & materials
MAN-PKR-4500	B	C	Sec 4 element selection guidance
Price book PB-25	-	-	Add kit SKU

4. Validation / Qualification Evidence

Aflas element package qualified by fixture test per API 11D1 V3 protocol at 350 degF / 10,000 psi differential, 3 thermal cycles (Test report TR-24-112). Immersion aging per NORSOK M-710 Annex A in 10% H2S / 20% CO2: volume swell 4.1%, hardness change +3 pts.

5. Disposition of Existing Stock and WIP

Existing stock: no action (std product unchanged). Kits built to order.

6. Implementation Checklist

Task	Owner	Status
Update drawings and BOM in PLM	Design Engineering	COMPLETE
Update datasheet and manual	Document Control	COMPLETE
Update router / work instructions	Manufacturing Engineering	COMPLETE

Task	Owner	Status
Train assembly & QC personnel	Quality	COMPLETE
Notify affected customers / bulletin	Sales Engineering	COMPLETE
First-article inspection on next lot	Quality	SCHEDULED

7. Approvals

Role	Name	Signature	Date
Engineering	J. Martinez, P.E.	/signed electronically/	2024-10-14
Quality	S. Nguyen	/signed electronically/	2024-10-15
Manufacturing	T. Okafor	/signed electronically/	2024-10-15
Doc Control	L. Hargrove	/signed electronically/	2024-10-15